

Worked Example 1: EOQ Solution

$$A = 3.2, D = 2,400/\text{yr}, v = 0.4, r = 0.24$$

Inventory and Production Management

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1. Model and formulas

Inputs

Ordering cost $A = 3.20/\text{order}$; demand $D = 2,400 \text{ units/yr}$; unit cost $v = 0.400$; carrying charge $r = 0.240 \text{ /yr}$.

Total relevant cost

$$TRC(Q) = \frac{Qvr}{2} + \frac{AD}{Q}$$

Optimal order quantity (Wilson formula)

$$EOQ = \sqrt{\frac{2AD}{vr}} = \sqrt{\frac{2(3.2)(2400)}{(0.4)(0.24)}} = 400.00$$

$$TRC(EOQ) = \sqrt{2ADvr} = 38.40 \text{ \$/yr}$$

Other expressions

$$T_{EOQ} = \sqrt{\frac{288A}{Dvr}} = 2.000 \text{ months}; \quad TR = \sqrt{\frac{2vrD}{A}} = 12.00$$

Sensitivity

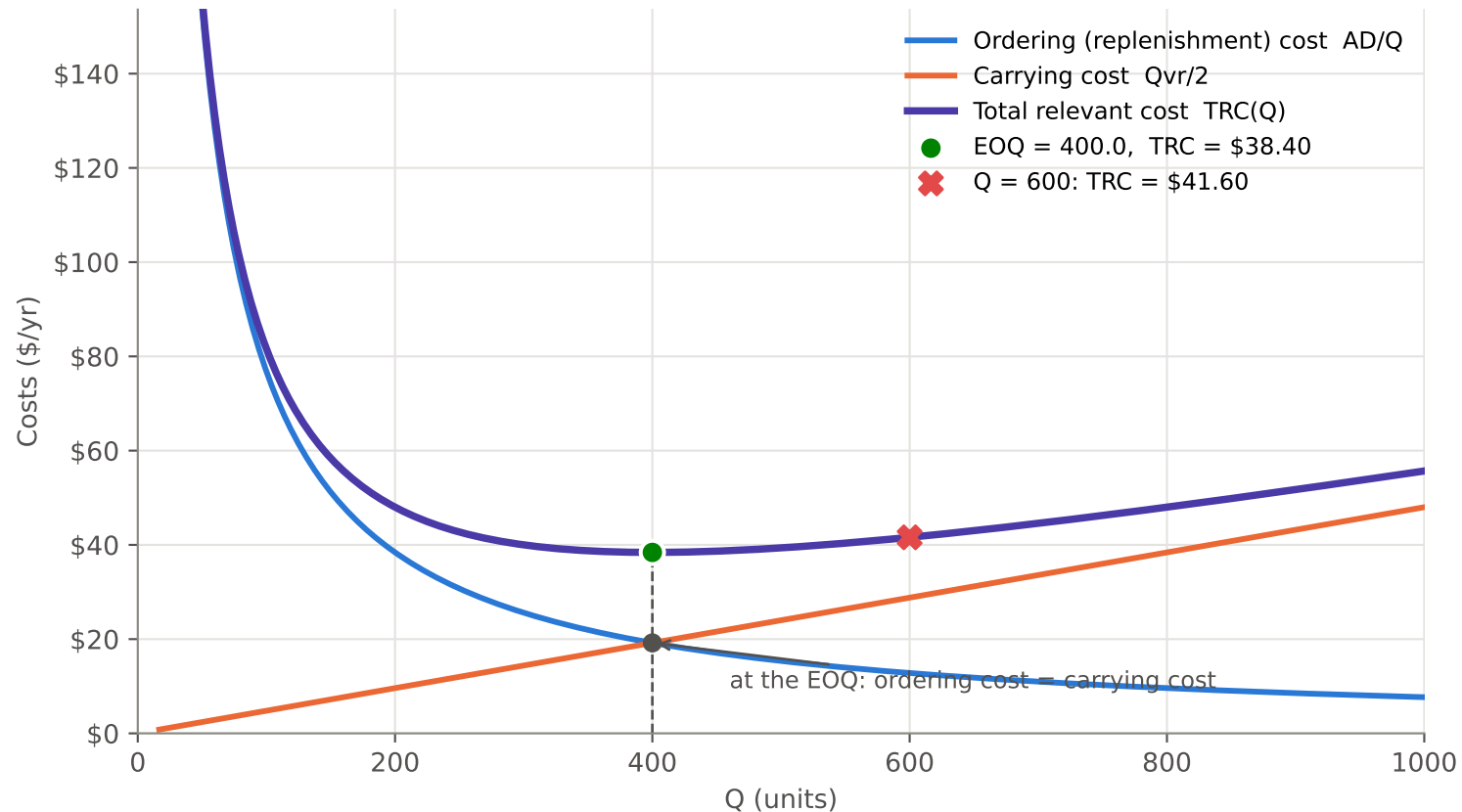
$$PCP = 50 \frac{\alpha^2}{1+\alpha}, \quad Q' = (1 + \alpha)EOQ$$

Using $Q' = 600$ ($\alpha = +0.500$) gives $PCP = 8.333\%$.

2. Numerical results

Quantity	Value
EOQ (units)	400
TRC(EOQ) (\$/yr)	38.4
Ordering cost AD/Q (\$/yr)	19.2
Carrying cost $Qvr/2$ (\$/yr)	19.2
Purchase cost vD (\$/yr)	960
Orders per year D/Q	6
Time between orders Q/D (years)	0.1667
T_{EOQ} (months of supply)	2
Time between orders (weeks)	8.667
Average inventory $Q/2$ (units)	200
Average inventory value (\$)	80
Turnover ratio $TR = D/I_{bar}$	12
Q used (units)	600
$\alpha = Q'/EOQ - 1$	0.5
TRC(Q used) (\$/yr)	41.6
PCP of Q used (%)	8.333

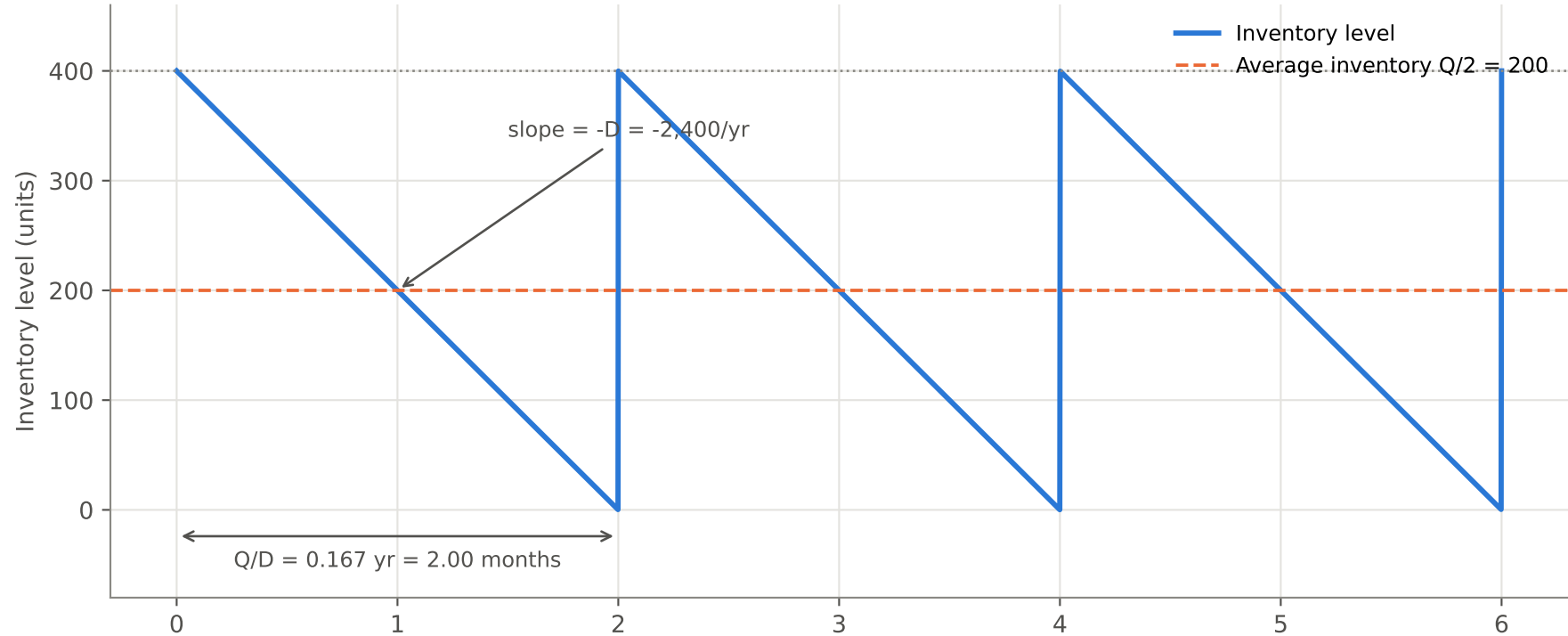
TRC(Q): EOQ = 400.0



Cost components (PCP of Q used = 8.33%)



Inventory profile at the EOQ



Percentage cost penalty curve

